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Effect of Cash Benefits on Health Care Utilization and Health
A Randomized Study

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IMPORTANCE Poverty is associated with greater barriers to health care and worse health outcomes, but it remains unclear whether income support can improve health.

OBJECTIVE To examine the effect of cash benefits on health care utilization and health.

DESIGN, SETTING, AND PARTICIPANTS The City of Chelsea, Massachusetts, a low-income community near Boston, randomly assigned individuals by lottery to receive cash benefits. Participants' medical records were linked across multiple health systems. Outcomes were assessed during the intervention period from November 24, 2020, to August 31, 2021.

INTERVENTION Cash benefits via debit card of up to $400 per month for 9 months.

MAIN OUTCOMES AND MEASURES The primary outcome was emergency department visits. Secondary outcomes included specific types of emergency department visits, outpatient use overall and by specialty, COVID-19 vaccination, and biomarkers such as cholesterol levels.

RESULTS Among 2880 individuals who applied for the lottery, mean age was 45.1 years and 77% were female. The 1746 participants randomized to receive the cash benefits had significantly fewer emergency department visits compared with the control group (217.1 vs 317.5 emergency department visits per 1000 persons; adjusted difference, −87.0 per 1000 persons [95% CI, −160.2 to −13.8]). This included reductions in emergency department visits related to behavioral health (−21.6 visits per 1000 persons [95% CI, −40.2 to −3.1]) and substance use (−12.8 visits per 1000 persons [95% CI, −25.0 to −0.6]) as well as those that resulted in a hospitalization (−27.3 visits per 1000 persons [95% CI, −53.6 to −1.1]). The cash benefit had no statistically significant effect on total outpatient visits (424.3 visits per 1000 persons [95% CI, −118.6 to 967.2]), visits to primary care (−90.4 visits per 1000 persons [95% CI, −308.1 to 127.2]), or outpatient behavioral health (83.5 visits per 1000 persons [95% CI, −182.9 to 349.9]). Outpatient visits to other subspecialties were higher in the cash benefit group compared with the control group (303.1 visits per 1000 persons [95% CI, 32.9 to 573.2]), particularly for individuals without a car. The cash benefit had no statistically significant effect on COVID-19 vaccination, blood pressure, body weight, glycated hemoglobin, or cholesterol level.

CONCLUSIONS AND RELEVANCE In this randomized study, individuals who received a cash benefit had significantly fewer emergency department visits, including those related to behavioral health and substance use, fewer admissions to the hospital from the emergency department, and increased use of outpatient subspecialty care. Study results suggest that policies that seek to alleviate poverty by providing income support may have important benefits for health and access to care.

Visual Abstract

Supplemental content

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Poverty is associated with greater barriers to health care and worse health outcomes.[1-7] Much of this relationship could be the result of confounding factors and reverse causality rather than income itself.[1] Therefore, whether income support can independently improve health remains an open question in the US.[2]

Income support administered as a cash benefit could, for example, reduce financial strain and improve mental health.[8] It could also help pay the costs associated with accessing health care. On the other hand, a common belief about cash benefits, supported by several cross-sectional and quasiexperimental studies, is that they enable misuse of substances and alcohol to the detriment of health.[9-16] Prior randomized studies of income support in the US have had modest to no effects on health.[17-28] Several of these studies, however, have been limited by small sample size or 1-time payments, and none have leveraged administrative health care data, which have precise records of health care utilization and biomarkers that can broaden the understanding of the effects of income. Depending on their effects on access and health, cash benefits could increase or decrease health care utilization.

From 2020 to 2021, the City of Chelsea, Massachusetts—a low-income, majority Latino and immigrant community near Boston—held a lottery to allocate monthly cash benefits to its low-income residents. Using the random assignment embedded in the lottery to overcome the issue of confounding, this study examines the impact of cash benefits on visits to the emergency department and outpatient care (overall and by specific categories of diagnoses and specialties), COVID-19 vaccination, and other markers of health measured in electronic health record data from hospitals and health systems in eastern Massachusetts.

Key Points

Question Among low-income individuals, how does a cash benefit affect use of the emergency department and outpatient care?

Findings In this study of 2880 randomized participants, there were significantly fewer emergency department visits among those assigned to receive a monthly cash benefit compared with the control group (217.1 vs 317.5 emergency department visits per 1000 persons), including fewer emergency department visits leading to hospital admission and fewer emergency department visits related to behavioral health and substance use. There were more outpatient visits to subspecialists, particularly for individuals without a car.

Meaning Cash benefits changed patterns of health care utilization in ways that suggest policies to alleviate poverty may improve health and access to health care.

Methods

Intervention, Randomization, and Study Population

The City of Chelsea was disproportionately affected by the health and economic consequences of the COVID-19 pandemic. By the end of 2020, the city had the highest COVID-19 cumulative incidence rate in Massachusetts.[29] Furthermore, the population of Chelsea was heavily concentrated in sectors of the economy that shut down, and its large undocumented population was not eligible for unemployment insurance or federal Economic Impact Payments.[30] After operating a large food distribution program from April through August 2020, the city decided to redirect its efforts toward providing direct financial support to its residents. The city accepted applications between July 27, 2020, and August 17, 2020, for Chelsea Eats debit cards (eFigure 1 in Supplement 2). Applicants with 1 person, 2 people, or at least 3 people in the family could receive $200, $300, or $400 per month, respectively.

To be eligible, individuals had to reside in Chelsea and have a family income below 30% of the median income for the Boston metropolitan statistical area (eMethods 1 in Supplement 2). Only 1 individual from each family was permitted to enter the lottery. Applicants could receive additional lottery tickets by meeting specific criteria, such as having a child or older adult in the family, being a veteran or disabled, receiving no unemployment benefits or having no one working in the family, or being ineligible for food assistance programs. The probability of winning the lottery was therefore higher among participants with more tickets (eTables 1 and 2 in Supplement 2). Randomization via the lottery occurred at the individual level and was held on September 17, 2020. Individuals randomly selected in the lottery could pick up their debit cards at Chelsea City Hall. The debit cards, first filled on November 24, 2020, and refilled monthly for 8 additional months, could be spent anywhere Visa was accepted.

Data

We obtained electronic health record data for 2019 through 2022 from 3 major health systems encompassing 15 hospitals, 30 health centers, and many additional outpatient clinics throughout the Boston area. These 3 health systems account for approximately 78% of the emergency, inpatient, and outpatient care among Chelsea residents (eMethods 2 in Supplement 2). Using application data for all lottery participants, we probabilistically linked the participants to their medical records at each of the health systems; 95% of the study population linked to at least 1 health system (eTable 3 in Supplement 2). The electronic health record data included detailed information on date and type of visit, department and specialty, vital signs, and diagnoses. It also included data on vaccinations and laboratory results. We separately collected survey data, described in more detail elsewhere,[31] from a subset of lottery participants, and used the baseline survey data to define subgroups of interest.

Our study period extends from November 24, 2020, the first date the debit cards were filled, through August 31, 2021, 4 weeks after the date the debit cards were refilled for the last time, representing an approximately 9-month observation period. The prerandomization period included January 1, 2019, the date our data begin, through September 16, 2020, the day before the lottery.

Outcome Measures

We prespecified 1 primary outcome, emergency department visits.[32] For our prespecified secondary outcomes, we assessed

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4 types of emergency department visits: those that resulted in admission to the hospital, emergency department visits related to behavioral health (which included mental illness and substance use disorder), emergency department visits related to substance use disorder, and potentially avoidable emergency department visits that represented visits that may have been treatable in less urgent settings. These visits were categorized using an updated version of the New York University algorithm.[33] We then measured the number of outpatient visits in total and by specialty, namely primary care, behavioral health, urgent care, and other subspecialties. Outpatient visits included office, virtual, and telemedicine visits. Additional prespecified secondary outcomes included receipt of at least 1 dose of any of the COVID-19 vaccines and several biomarkers—systolic and diastolic blood pressure, body weight, glycated hemoglobin, low-density lipoprotein cholesterol, and total cholesterol. Because the biomarkers were observed only for those who used medical care, which itself could be influenced by the cash benefit, we measured the probability of having a reading for each of the biomarkers, and the analytical sample for the biomarkers was limited to individuals who had readings in both the pre- and postintervention periods. More details about the outcomes are included in the supplementary material, including the classification of emergency department visits and post hoc outcomes (eMethods 3 in Supplement 2).

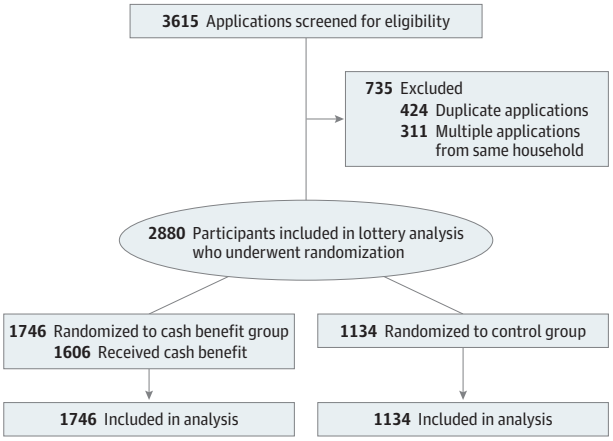
Statistical Analysis

The protocol (Supplement 1), which included our prespecified analysis plan and outcomes, was published before analysis of the data (NCT05622903).[32] We fit a linear regression model in an intention-to-treat analysis to compare outcomes for individuals randomized into the cash benefit group with individuals randomized into the control group. Because randomization was conditional on the number of lottery tickets, we weighted participants in the control group by the probability of winning the lottery divided by the probability of not winning the lottery (eMethods 4, eTable 2 in Supplement 2).[34] To increase precision, we adjusted for age, sex, and baseline versions of the outcome from the prerandomization period. We conducted sensitivity analyses to ensure our results were robust to alternative specifications, such as controlling for the number of tickets (eMethods 5 in Supplement 2).

We conducted 4 prespecified and 2 post hoc subgroup analyses (eMethods 6 in Supplement 2). Two subgroups—those with prior acute care utilization and chronic disease at baseline—were defined from the full sample based on electronic health record data. Four additional subgroups—poor self-reported health, financial distress, severe psychological distress (post hoc), and lack of car ownership (post hoc)—were defined using baseline survey data from the subset of the full sample that participated in the survey. To explore whether the survey subsample was representative of the full study population, we separately assessed our results by survey participation.

All statistical tests were 2-tailed with a level of significance set at $P < .05$. We reported 95% CIs and standard P values for all outcomes. Because of the potential for type I error due to multiple comparisons, results from our secondary outcomes should be considered exploratory; to aid interpretation of our analyses, we conducted supplemental significance tests using the Benjamini-Hochberg procedure as well as omnibus F-tests for each group of secondary outcomes. Analyses were conducted using Stata, version 17.0 (StataCorp). This study was approved with a waiver of informed consent by the institutional review boards at Mass General Brigham and Cambridge Health Alliance. This study followed the Consolidated Standards of Reporting Trials (CONSORT) reporting guidelines.

Figure. Participants in a Study of Cash Benefits and Health Care Utilization



Results

Study Population

There were 1746 individuals randomized to the cash benefit group and 1134 individuals randomized to the control group (**Figure**). The study population was 77% female. The mean family size was 3.2, and the mean annual family income was $16 709. Characteristics (**Table 1**) and distance to nearby hospitals (eFigure 2, eTable 4 in Supplement 2) were balanced between the cash benefit and control groups. Of cash benefit group participants, 1606 (92%) picked up their debit card. Card recipients spent a mean of $335 per month, which represented, on average, a 23% increase in baseline monthly income.

Emergency Department Use

Table 2 shows the estimated effects of the cash benefit on emergency department use. During the 9-month intervention period, there were 217.1 emergency department visits per 1000 persons in the cash benefit group and 317.5 emergency department visits per 1000 persons in the control group. The adjusted difference was 87.0 fewer visits per 1000 persons (95% CI, −160.2 to −13.8; $P = .02$), representing a relative decrease of 27%. Similarly, 14.0% of individuals in the cash benefit group and 18.1% of individuals in the control group had any

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Table 1. Baseline Characteristics of the Study Population

Characteristics[a]	Cash benefit group (n = 1746)	Control group (n = 1134)	Difference (95% CI)[b]
Sex, %	n = 1743	n = 1128	
Female	76.4	76.7	−0.3 (−3.9 to 3.3)
Male	23.6	23.3	0.3 (−3.3 to 3.9)
Age categories, %	n = 1689	n = 1093	
≤34 y	28.2	25.9	2.3 (−1.2 to 5.8)
35-49 y	40.5	41.6	−1.1 (−5.2 to 3.0)
≥50 y	31.3	32.5	−1.2 (−5.3 to 2.9)
Age, mean (SD), y	44.6 (14.5)	45.6 (15.5)	−0.9 (−2.4 to 0.5)
With disability, %	22.8	22.6	0.2 (−3.7 to 4.1)
Veteran, %	2.3	3.5	−1.2 (−3.6 to 1.3)
Spanish as preferred language, %	74.6	72.1	2.5 (−1.3 to 6.3)
Diagnosed with depression or anxiety, %	25.4	24.8	0.5 (−3.0 to 4.1)
Any emergency department use, %	28.4	30.8	−2.5 (−6.3 to 1.4)
Any outpatient use, %	81.7	79.9	1.8 (−1.5 to 5.1)
Any primary care visit, %	71.7	70.4	1.3 (−2.5 to 5.1)
Characteristics of applicant's family			
Family income categories, $			
0 to <10 000	30.5	32.5	−2.0 (−5.9 to 1.8)
10 000 to <24 000	35.1	36.1	−1.0 (−5.1 to 3.0)
≥24 000	34.5	31.4	3.1 (−0.6 to 6.8)
Family mean income (annual), $	17 154.3	16 264.6	889.7 (−100.3 to 1879.7)
Family size categories, %[c]			
1 person	18.7	19.7	−1.0 (−4.5 to 2.5)
2-4 people	59.7	60.0	−0.2 (−4.3 to 3.9)
≥5 people	21.6	20.4	1.2 (−2.1 to 4.6)
Family size, mean[c]	3.2	3.2	0.0 (−0.1 to 0.2)
No. of children in family, mean, %	1.5	1.4	0.0 (−0.1 to 0.2)
At least 1 child aged 0-5 y	39.9	38.9	1.0 (−3.0 to 5.1)
At least 1 child aged 6-17 y	57.8	58.0	−0.2 (−4.2 to 3.9)
At least 1 older adult aged ≥65 y, %	22.3	21.9	0.4 (−3.6 to 4.4)
At least 1 family member working, %	42.1	40.3	1.8 (−2.1 to 5.7)
Receiving unemployment assistance, %	17.8	18.9	−1.2 (−4.1 to 1.7)
Receiving food assistance (eg, SNAP), %	41.6	42.1	−0.4 (−4.4 to 3.5)

Abbreviation: SNAP, Supplemental Nutrition Assistance Program.

[a] Family characteristics, disability status, veteran status, and preferred language came from the lottery application form (eFigure 1 in Supplement 2). Other characteristics, including age, sex, mental health diagnosis, and prior utilization, came from the electronic health records. All estimates are weighted as described in the text to account for the number of lottery tickets. Baseline characteristics for diagnoses and utilization were assessed before the lottery from January 1, 2019, through September 16, 2020.

[b] For variables expressed as percentages, the difference is expressed as percentage points.

[c] Family size represented number of family members living at the same address as the applicant.

Table 2. Absolute Change in Emergency Department Use

Utilization measure	Mean visits per 1000 persons		Difference (95% CI)		*P* value for adjusted difference
	Cash benefit group	Control group	Unadjusted	Adjusted[a]	
Emergency department use	217.1	317.5	−100.4 (−179.3 to −21.6)	−87.0 (−160.2 to −13.8)	.02
Emergency department use by type					
Admission to hospital[b]	36.1	65.7	−29.6 (−56.9 to −2.4)	−27.3 (−53.6 to −1.1)	.04
Behavioral health related[c,d]	7.4	34.6	−27.2 (−50.4 to −3.9)	−21.6 (−40.2 to −3.1)	.02
Substance use disorder related[c,e]	1.7	14.7	−13.0 (−25.9 to −0.0)	−12.8 (−25.0 to −0.6)	.04
Potentially avoidable[c,f]	78.0	94.8	−16.7 (−45.5 to 12.1)	−10.2 (−38.2 to 17.9)	.48

[a] Estimates were adjusted for sex, age in categories, and baseline utilization.

[b] Emergency department use with admission to hospital represented emergency department visits that resulted in hospitalization based on disposition from the emergency department and encounter category in the electronic health records.

[c] Emergency department visits are classified following Johnston et al.[33]

[d] Behavioral health–related emergency department use represented emergency department visits that, based on the primary diagnosis for the visit, were mental health, alcohol, or drug related.

[e] Substance use–related emergency department use was a subset of behavioral health–related emergency department visits and represented emergency department visits that, based on the primary diagnosis for the visit, were alcohol or drug related.

[f] Potentially avoidable emergency department use represented emergency department visits that were nonemergent or treatable through primary care based on the primary diagnosis for the visit.

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Table 3. Absolute Change in Outpatient Use

Utilization measure	Mean visits per 1000 persons		Difference (95% CI)		*P* value for adjusted difference
	Cash benefit group	Control group	Unadjusted	Adjusted[a]	
Outpatient use[b]	4778.4	4313.6	464.8 (−113.3 to 1042.8)	424.3 (−118.6 to 967.2)	.13
Outpatient use by specialty[c]					
Primary care	1805.3	1887.9	−82.6 (−323.8 to 158.6)	−90.4 (−308.1 to 127.2)	.42
Behavioral health	769.2	629.5	139.7 (−147.5 to 426.8)	83.5 (−182.9 to 349.9)	.54
Urgent care	441.0	371.4	69.6 (−6.1 to 145.3)	68.8 (−3.3 to 140.8)	.06
Other subspecialties	1762.9	1424.8	338.1 (54.4 to 621.8)	303.1 (32.9 to 573.2)	.03

[a] Estimates were adjusted for sex, age in categories, and baseline utilization.

[b] Outpatient use included office, virtual, and telemedicine visits based on encounter category in the electronic health records.

[c] Outpatient use by specialty included office, virtual, and telemedicine visits to primary care, behavioral health, urgent care, or other subspecialties based on the department and specialty categories in the electronic health records.

emergency department visit (adjusted difference, −3.4 percentage points [95% CI, −6.5 to −0.3]). There was no difference in emergency department visits during the anticipation period between the lottery and first disbursement (eTable 5 in Supplement 2). During the 9 months after discontinuation of the program, the estimate for the difference between the cash benefit and control groups was slightly smaller than during the 9-month intervention period, but the 95% CI includes 0 (eTable 5 in Supplement 2).

The reduction in emergency department visits varied by type. The largest relative decreases were among behavioral health-related emergency department visits (−21.6 visits per 1000 persons [95% CI, −40.2 to −3.1]), representing a decrease of 62% relative to the control mean, and substance use-related emergency department visits (−12.8 visits per 1000 persons [95% CI, −25.0 to −0.6]), representing a relative decrease of 87%. There were also reductions in emergency department visits that resulted in a hospitalization (−27.3 visits per 1000 persons [95% CI, −53.6 to −1.1]), a relative decrease of 42%. The cash benefit had no statistically significant effect on potentially avoidable emergency department visits (−10.2 visits per 1000 persons [95% CI, −38.2 to 17.9]).

Outpatient Use

Table 3 shows the estimated effects of the cash benefit on outpatient use. There were 4313.6 outpatient visits per 1000 persons in the control group during the 9-month intervention period. The number of outpatient visits was higher in the cash benefit group by 424.3 visits per 1000 persons (95% CI, −118.6 to 967.2); this adjusted difference was not statistically significant (Table 3).

Differences in outpatient use varied across specialties. There was no statistically significant difference in outpatient visits to primary care (−90.4 visits per 1000 persons [95% CI, −308.1 to 127.2]), outpatient behavioral health (83.5 visits per 1000 persons [95% CI, −182.9 to 349.9]), or urgent care (68.8 visits per 1000 persons [95% CI, −3.3 to 140.8]). However, there was an increase in visits to other subspecialties (303.1 visits per 1000 persons [95% CI, 32.9 to 573.2]). There were similar increases across a variety of medical, surgical, ancillary, and specific subspecialties that were not statistically significant individually (eTable 6 in Supplement 2). There was a larger increase in visits to subspecialty clinics further from Chelsea where participants lived (eTable 6 in Supplement 2). The cash benefit had no statistically significant effect on prescriptions ordered in the electronic health record, including antidepressants or substance use treatment (eTable 6 in Supplement 2).

COVID-19 Vaccination and Biomarkers

The COVID-19 vaccination rate in the control group was 84.2%, and the cash benefit had no statistically significant effect on COVID-19 vaccination rates (1.1 percentage points [95% CI, −1.9 to 4.1]) (**Table 4**). In the control group among those with readings in the pre- and post-intervention periods, the mean systolic blood pressure was 124.3 mm Hg, mean body weight was 78.0 kg, mean glycated hemoglobin was 7.5%, and mean low-density lipoprotein cholesterol was 104.1 mg/dL. The cash benefit had no statistically significant effect on the biomarkers (Table 4) or the probability of having a reading (eTable 7 in Supplement 2).

Supplemental Analyses

The results from the main analyses were robust to a variety of sensitivity analyses, including analyses that controlled for the number of lottery tickets or a larger set of baseline characteristics (eTable 8 in Supplement 2). After accounting for testing of multiple hypotheses, the estimates for the subtypes of emergency department use that were significant remained statistically significant at a false discovery rate of 10%, but the estimate for outpatient use to other specialties did not (eTable 9 in Supplement 2). An F-test rejects the null of no overall effect among the subtypes of emergency department use and among the outpatient outcomes.

As discussed earlier, we defined several subgroups using baseline survey data collected from a subset of the study population; 57% of individuals participated in the survey and 43% did not. Because our electronic health record data included both survey participants and survey nonparticipants, we first compared how the baseline characteristics (eTable 10 in Supplement 2) and results (**Table 5**; eTables 11 and 12 in Supplement 2) in these subsamples differed. Emergency department visits were concentrated among those who did not participate in the survey, and there was a statistically significant reduction in this outcome in the cash benefit group compared with the control group for survey nonparticipants (−149.0 visits per 1000 persons [95% CI, −290.4 to −7.6]). The reduction in emergency department use for survey participants was smaller and not statistically significant. In contrast, the impact on outpatient use was concentrated among the survey participants, for

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Table 4. Absolute Change in COVID-19 Vaccination and Clinical Measures

Variable	Cash benefit group		Control group		Difference (95% CI)		*P* value for adjusted difference
	No. (n = 1746)[a]	Mean	No. (n = 1134)[a]	Mean	Unadjusted	Adjusted[b]	
COVID-19 vaccination, %	1489	85.0	987	84.2	0.8[c] (−2.2 to 3.8)	1.1[c] (−1.9 to 4.1)	.49
Systolic blood pressure, mm Hg	1071	124.6	674	124.3	0.3 (−2.0 to 2.7)	0.4 (−1.5 to 2.3)	.68
Diastolic blood pressure, mm Hg	1071	74.5	674	74.1	0.4 (−0.7 to 1.4)	0.4 (−0.5 to 1.4)	.37
Weight, kg	948	80.6	597	78.0	2.6 (0.6 to 4.6)	0.3 (−0.3 to 1.0)	.33
Hemoglobin A_{1c}, %	380	7.0	234	7.5	−0.5[c] (−1.0 to −0.1)	−0.3[c] (−0.6 to 0.1)	.17
Low-density lipoprotein cholesterol, mg/dL	181	95.4	112	104.1	−8.7 (−18.9 to 1.5)	−7.5 (−15.5 to 0.5)	.07
Total cholesterol, mg/dL	189	180.7	116	186.8	−6.1 (−18.5 to 6.2)	−5.4 (−15.1 to 4.4)	.28

[a] The sample size for the COVID-19 vaccination outcome consisted of individuals for whom vaccination records were available, including those whose electronic medical records were connected to the statewide immunization registry, and the sample size for the clinical measures consisted of the subpopulation with readings in the pre- and postintervention periods; see main text for details.

[b] Estimates were adjusted for sex, age in categories, and baseline values of the outcome (except for the COVID-19 vaccination measure).

[c] For variables expressed as percentages, the difference is expressed as percentage points.

Table 5. Comparing Results From Administrative Data Based on Survey Participation

Utilization measure	In survey sample (n = 1643)			Not in survey sample (n = 1237)		
	Control group, mean visits per 1000 persons	Effect of cash benefit (95% CI)[a]	*P* value	Control group, mean visits per 1000 persons	Effect of cash benefit (95% CI)[a]	*P* value
Emergency department use (overall and by type)						
Overall	231.2	−43.1 (−114.5 to 28.4)	.24	433.5	−149.0 (−290.4 to −7.6)	.04
Hospitalized	36.8	−8.0 (−29.5 to 13.5)	.47	104.5	−52.8 (−105.3 to −0.3)	.05
Behavioral health[b]	10.8	−4.4 (−14.1 to 5.4)	.38	66.7	−39.8 (−77.3 to −2.2)	.04
Substance use	2.9	−2.1 (−6.5 to 2.2)	.33	30.6	−24.2 (−48.6 to 0.1)	.05
Potentially avoidable	79.1	−8.7 (−40.7 to 23.4)	.60	115.9	−16.6 (−67.8 to 34.5)	.52
Outpatient use (overall and by specialty)						
Overall	4359.6	718.0 (−73.8 to 1509.8)	.08	4251.7	57.1 (−667.0 to 781.2)	.88
Primary care	1910.5	−80.7 (−382.7 to 221.3)	.60	1857.4	−102.3 (−415.4 to 210.7)	.52
Behavioral health[b]	696.6	257.1 (−161.9 to 676.1)	.23	539.3	−111.8 (−394.8 to 171.3)	.44
Urgent care	408.0	45.1 (−42.0 to 132.2)	.31	322.2	97.3 (−26.3 to 220.8)	.12
Other subspecialties	1344.4	520.8 (152.5 to 889.1)	.01	1532.8	30.4 (−371.2 to 431.9)	.88

[a] Estimates were adjusted for sex, age in categories, and baseline utilization. Unadjusted estimates are available in eTables 11 and 12 in Supplement 2.

[b] Emergency department visits related to behavioral health were defined by the primary diagnosis of the emergency department visit, whereas outpatient use of behavioral health was defined by the specialty of the outpatient visit.

whom there was an increase in visits to other subspecialties (520.8 visits per 1000 persons [95% CI, 152.5 to 889.1]). Within the subsample of survey participants, this increase in outpatient visits to other subspecialties was larger for individuals who had no car ownership, poor health, severe psychological distress, and financial distress at baseline (eTable 13 in Supplement 2). In subgroups defined using the electronic health records, specifically individuals with a history of acute care utilization and chronic disease, there was an increase in outpatient visits to other subspecialties and decrease in emergency department visits related to behavioral health, larger than that in the full study population (eTable 14 in Supplement 2).

Discussion

This randomized study found that individuals who received a monthly cash benefit had significantly fewer visits to the emergency department, including those related to behavioral health and those resulting in admission to the hospital. Compared with prior studies of income and health, this study had several advantages that enabled the detection of meaningful effects. The cash benefits administered by the City of Chelsea were recurring, unconditional, unrestricted, large relative to baseline income, and randomized at scale. Another key feature of the study was the use of administrative health data, which have been underutilized in the study of the income-health relationship due to the lack of detailed socioeconomic measures in health care records and the fragmentation of care across insurers and health providers. These challenges were overcome by making use of randomization of cash benefits and linking participants' administrative data across health systems. The benefits of such administrative data include longitudinal follow-up, less risk for misclassification or response bias, an ability to capture clinical details not always possible via self-report, and reduced susceptibility to nonresponse bias.[35,36] Administrative data captured

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the health care utilization of those who were less likely to participate in surveys, a group in which emergency department visits, particularly those related to behavioral health, were concentrated. To the extent that survey-based studies implicitly overlook individuals with severe mental illness, this may help explain the modest to null findings of prior randomized studies of cash benefits.[17-28,31] It could also have implications for the use of national surveys in the evaluation of social policy amid a 3-decade decline in response rates.[37-39]

How did the cash benefit reduce emergency department utilization? Given that the study saw less use of the emergency department and greater use of outpatient care (for subspecialists), 1 interpretation of these results is a replacement of emergency care with outpatient care. However, this is unlikely to be the main explanation for these findings. There was no significant effect of the cash benefit on prescriptions or on outpatient visits to primary care, urgent care, and behavioral health, which would more directly substitute for emergency department visits compared with outpatient visits to other subspecialties. Furthermore, the effects of the cash benefit on acute and outpatient care appear in 2 distinct groups of people, identified by their participation or nonparticipation in the survey. The decrease in visits to the emergency department was concentrated in the subgroup who did not participate in the survey and represents an older, more disabled population with less income, whereas the increase in visits to outpatient subspecialists was concentrated in the subgroup that participated in the survey and represents a younger, less disabled population with higher baseline income.

A more likely explanation for the fewer visits to the emergency department is a direct improvement in health from the cash benefit that reduced the demand or need for emergency care. Financial strain is associated with reduced cognitive bandwidth, more mental illness, and greater use of alcohol and other substances.[40-42] The cash benefit, by reducing financial strain and improving economic resilience, may have had direct positive effects on the mental health of lower-income individuals. This is consistent with randomized studies of cash transfers in low- and middle-income countries,[43-45] as well as quasiexperimental evidence in the US from the effect of the Earned Income Tax Credit and Child Tax Credit on survey measures of mental health.[46-49] An improvement in mental health could, in turn, have led to the fewer emergency department visits related to behavioral health that the study observed in the cash benefit group.

The study also found fewer emergency department visits related to substance use in the cash benefit group, which contradicts previous study results suggesting that cash transfers increase alcohol- or substance-related morbidity and mortality.[9-16] These contrasting results can potentially be explained by differences in methodology and time frame. Specifically, several of the prior studies leverage quasirandomization in the day of the month that a cash transfer arrives and these fluctuations in income may result in transient effects over days to weeks rather than an absolute change over a longer time horizon.[16,50] In contrast, this study leverages random allocation of cash benefits via a lottery, and the outcomes were assessed over a 9-month period.

The cash benefit did not change utilization of primary care or COVID-19 vaccination rates. This may have been because primary care was already readily accessible in Chelsea, and during the study period, the COVID-19 vaccine was available for free at a local mass vaccination site. Instead, the cash benefit increased use of subspecialty care, which is less accessible in Chelsea and often requires a more burdensome trip into downtown Boston. Transportation is a well-known barrier to care, and the cash benefit may have helped overcome this barrier.[51,52] Consistent with this mechanism, the study observed larger effects for visits to clinics further from Chelsea and for those without a car. These findings suggest that, for lower-income individuals, access to health care can be improved by making health care more reachable locally with low to no out-of-pocket costs, as well as by increasing patients' discretionary income to overcome logistics related to travel and time.[53]

By decreasing the demand for more expensive acute care relative to outpatient care, cash benefits have the potential to be cost saving to the health care system. The electronic health record data lacked corresponding cost or billing data, so the study was unable to estimate effects by insurance status or on total health care spending. However, assuming $16 903 for the typical cost of a hospitalization,[54] $757 for an emergency department visit,[54] and $122 for an outpatient visit,[55] the cash benefit could have resulted in net savings to the overall health care system of approximately $450 per person over 9 months. These savings would cover approximately one-sixth to one-seventh of the debit cards' costs and could be as much as half of the debit cards' total costs to the extent that some benefits accrue to other members of the family (given a mean family size of 3 in the study) or persist beyond 9 months.

Limitations

Estimates of the effects of cash benefits apply most directly to low-income families with children and working-age parents, a population of considerable policy interest given ongoing debates about expanding the Child Tax Credit into a monthly child allowance program, as well as recent guaranteed income pilots in cities across the US.[56] However, there are important limitations to the generalizability of study findings. First, the low-income population of Chelsea, Massachusetts, differs from the overall low-income population of the US. For example, the City of Chelsea has a higher proportion of residents born outside the US and a lower uninsured rate. Second, the intervention occurred during the early years of the COVID-19 pandemic, a unique period in terms of the economy and public health. Nonetheless, nearly 30% of US families live below 200% of the federal poverty level and likely face many of the same economic constraints and barriers to health care as the individuals in this study. Third, the study was unable to access care information outside of the 3 health systems in the data. Statewide registry data, however, confirmed that the study likely captured a substantial portion of health care utilization for individuals in the study. If the cash benefit group was more inclined to shift care to health care providers outside of the dataset, estimates for the effect of cash benefits on emergency

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department use may be modestly overstated, but differential transfer of care is likely minimal; study data show that 98% of individuals who sought primary care and 88% of individuals who sought emergency care did so within the same health system in the period before vs after the study, with no difference between the cash benefit and control groups. Fourth, although the study did not detect any significant effect on biomarkers, power was limited by the small number of patients who had hypertension, hyperlipidemia, or diabetes; the effects of income may also differ if the cash benefits were extended for a longer duration.

Conclusions

In this randomized study, individuals who received a cash benefit had significantly less use of the emergency department, particularly for reasons related to behavioral health and substance use, and decreased admissions to the hospital from the emergency department. They also had higher use of outpatient subspecialty care. Policies that alleviate poverty by providing income support in the form of cash benefits may produce important benefits for health and access to health care.

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